

BOUDROUSS RÉDA | CV

AI Engineer – LLM Agents, Verification & Static Analysis – Paris

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Languages: French (Native), English (Professional), Arabic (Conversational)

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Profile

At Hexaglobe I industrialized LLM-agent test generation with sandboxed **multi-agent orchestration**, versioned **guardrails**, mandatory green tests and systematic human review, so that no AI output is trusted without **verification**. I also bring a **formal-methods** background from Sorbonne University, where I studied **abstract interpretation** and program verification, ported the **MOPSA** analyzer (OCaml/C/C++) to WebAssembly and wrote an abstract-interpretation-based analyzer in **Rust**. I would like to combine both sides at TrustInSoft, backing AI productivity with sound analysis.

Experience

Quality Analyst Engineer (apprenticeship) - Hexaglobe (video streaming)

2025 – 2026

- **Industrialized AI-assisted development** in an enterprise environment, building a specification-to-execution loop (PRDs, issues, **multi-agent orchestration** in Docker sandboxes with planner, implementer and reviewer agents) that produced **~720 integration tests in 3 days** under systematic human review.
- Designed the **prompting strategies and guardrails** that make delegation reliable. Every AI contribution has to pass mandatory green tests, follow versioned and codified agent rules, and stay fully traceable.
- Built the verification foundation first, writing **~1,400 test cases** by hand (Playwright, Jest, pytest, PHPUnit) for zero-coverage legacy products across 4 stacks (TypeScript, Python, PHP, JS).
- Developed objective **quality oracles** (VMAF, black-frame/freeze detection, audio analysis) to validate a live video encoder, plus shared **GitLab CI/CD** components (centralized Allure reporting) consumed by 4+ repositories.

Technical Lead & President (2024–25) - Association Play Sorbonne Université

2023 – 2026

- Designed and operated a virtualized **infrastructure** (Docker servers, CI/CD, networking). As president, led **100+ volunteers** for a cultural event welcoming 10,000+ visitors.

Projects

- **Mopsa-wasm**: ported **MOPSA**, a sound **abstract-interpretation** static analyzer for C and Python developed at LIP6 (OCaml/C/C++), to **WebAssembly**, with a fully client-side **playground** and a technical **write-up**.
- **Reactant analyzer**: an abstract-interpretation-based static analyzer written in **Rust** for React web applications.

Education

MSc in Computer Science (Software Science & Technology) - Sorbonne University

2024 – 2026

- Coursework: **static analysis by abstract interpretation** and type systems (course led by Antoine Miné, MOPSA project lead), program specification & **formal verification**, inference of numerical properties, compiler construction, concurrent programming.

BSc in Computer Science - Sorbonne University

2020 – 2024

- Algorithms, computer architecture, operating systems, networks, databases, with team projects in C, Java, OCaml, Python and TypeScript.

Skills

AI engineering: LLM agent orchestration (Claude Code), prompting strategies, guardrail design, sandboxed agent execution (Docker)

Languages: C, OCaml, Rust, Python, TypeScript, PHP, SQL

Topics: abstract interpretation, testing (unit / integration / E2E), GitLab CI/CD, WebAssembly & Emscripten